

CALCULATION OF MULTI-INSTANTON DETERMINANTS

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The previous analysis of the variation of multi-instanton determinants for the covariant laplacian D^2 is extended to express $\delta \ln \det(-D^2)$ as δ of a 4-dimensional integral, where δ denotes an infinitesimal change in the instanton parameters, apart from a contribution involving the 4-dimensional antisymmetric symbol. The result, which gives an expression for the determinant up to a factor, extends the known result for the 't Hooft solution.

1. Introduction

In a previous paper by the present author and Corrigan, Goddard and Templeton [1], an investigation into the determinants that arise in the functional integration of fluctuations about general multi-instanton solutions for gauge theories on 4-dimensional euclidean space was initiated. Relying on the zeta-function method of regularisation and the knowledge of the Green functions for the general instanton solution * an expression for the variation of the basic determinant in the form of an integral over the variation of the gauge field due to a change in the parameters of the solution was obtained. In this paper, which is designed as a sequel to ref. [1] and relies heavily on it for notation and background, we show how to express this result as the corresponding variation of a four-dimensional integral over a function of the parameters of the general self-dual gauge field. This had previously been achieved only for the restricted 't Hooft solution [2], and the integral gives the complete determinant up to a factor independent of the parameters.

In the construction of the general k -instanton solution for self-dual gauge fields due to Atiyah, Hitchin, Drinfeld and Manin [3] (ADHM) a fundamental role is played by the linear operator $\Delta(x) = a + bx$ which maps from a $2k$ -dimensional linear space $\mathcal{W}$ (which is a tensor product of a k -dimensional real or complex space W , according to whether the gauge group is $\text{Sp}(r)$ or $\text{SU}(n)$, and a two-dimensional

* See ref. [1] for a complete listing of references.

space on which the quaternion x , in which is encoded the euclidean position coordinates, acts) to a $2k + 2r$ or $2k + n$ (for gauge groups $\text{Sp}(r)$ or $\text{SU}(n)$) dimensional linear space V . In ref. [1], and also a recent paper by Belavin, Fateev, Schwarz and Tyupkin [4], an expression for the variation of $\ln \det(-D^2)$, where $D_\mu = \partial_\mu + A_\mu$ is the covariant derivative containing the general instanton gauge field A_μ , was found in terms of the naturally arising projection operator P acting on the space V . However, it did not seem possible to undo the variation explicitly. Here we work in terms of the space $\mathcal{W}$ by going over a slightly modified formalism from ADHM due to Drinfeld and Manin [5] which then allows a more complete analysis of the problem of removing the variation.

In sect. 2 we therefore show how to re-express the results of ref. [1] in terms of just square $2k \times 2k$ dimensional matrices acting on the space $\mathcal{W}$ and how to obtain by partial integration a form in which the variation can be taken outside the integration. In sect. 3 we show how this can be specialised to the 't Hooft solution and re-derive the results for the conformal variation, previously obtained in ref. [1], sect. 4. The conclusion contains further remarks concerning a difficult term that arose in the calculations.

2. Variation of the determinant

The zeta-function method of regularisation, described in ref. [1], defines

$$\det\left(-\frac{D^2}{\mu^2}\right) = \exp[-\ln \mu^2 \zeta(0) - \zeta'(0)] . \quad (2.1)$$

In ref. [1] it was shown that for a self-dual gauge field A_μ of instanton number k $\zeta(0) = -\frac{1}{12}k$, and under a variation of the parameters in the general solution A_μ

$$\delta \zeta'(0) = \frac{1}{12\pi^2} \int d^4x \operatorname{tr} [\delta A_\mu J_\mu] . \quad (2.2)$$

A_μ is taken to be given by the ADHM construction,

$$A_\mu = V^\dagger \partial_\mu V , \quad (2.3)$$

while, from ref. [1] with the same notation,

$$J_\mu = V^\dagger b f (e_\mu \Delta^\dagger b - b^\dagger \Delta \bar{e}_\mu) f b^\dagger V , \quad (2.4)$$

is essentially the vacuum polarisation current due to the presence of the instanton gauge field ($e_\mu = \partial_\mu x$ forms a quaternionic basis, $\bar{e}_\mu$ is its conjugate).

Our analysis of (2.2) follows Drinfeld and Manin [5] in postulating an explicit form for

$$\Delta(x) = a + bx = \begin{pmatrix} \lambda^\dagger \\ B + Cx \end{pmatrix} , \quad (2.5)$$

where B, C are square $2k \times 2k$ dimensional matrices (in some contexts it is perhaps natural to take $C = -1$ but this restriction is preferably avoided here) while $\lambda^\dagger$ is a matrix acting from the space $\mathcal{W}$ to the G ($=n$ or $2r$ for $SU(n)$ or $Sp(r)$) dimensional representation space of the gauge group. The form of (2.5) enables a solution for $V^\dagger$ from $V^\dagger \Delta = 0$

$$V^\dagger = U^\dagger(1_N, -u^\dagger), \quad u(x)^\dagger = \lambda^\dagger(B + Cx)^{-1}, \quad (2.6)$$

where U acting on the group space ensures the normalisation condition $V^\dagger V = 1_N$

$$1_N + u^\dagger u = (UU^\dagger)^{-1}. \quad (2.7)$$

With (2.6), then from (2.3) and (2.7)

$$A_\mu = -U^\dagger \lambda^\dagger G^{-1} \bar{e}_\mu C^\dagger u U + U^{-1} \partial_\mu U, \quad (2.8)$$

for $G(x) = (B + Cx)^\dagger(B + Cx)$. The form of (2.8) suggests the gauge transformation

$$A_\mu = U^{-1} \tilde{A}_\mu U + U^{-1} \partial_\mu U, \quad (2.9)$$

so that

$$\tilde{A}_\mu = -UU^\dagger \lambda^\dagger G^{-1} \bar{e}_\mu C^\dagger u, \quad (2.10)$$

although as U is not unitary manifest hermiticity is lost. Now in the ADHM solution a fundamental role is played by the $k \times k$ matrix f , acting on W , which is initially defined through the critical constraint

$$\Delta^\dagger \Delta = 1_2 f^{-1}. \quad (2.11)$$

From (2.5)

$$\Delta^\dagger \Delta = \lambda \lambda^\dagger + G, \quad (2.12)$$

and as then $(UU^\dagger)^{-1} \lambda^\dagger = \lambda^\dagger G^{-1} f^{-1}$, (2.10) becomes

$$\tilde{A}_\mu = -\lambda^\dagger f \bar{e}_\mu C^\dagger u. \quad (2.13)$$

Further useful definitions are

$$\begin{aligned} P(x) &= b^\dagger \Delta(x) = C^\dagger(B + Cx), \\ P^\dagger &= \Delta^\dagger b, \\ b^\dagger b &= C^\dagger C = 1_2 \nu, \end{aligned} \quad (2.14)$$

so

$$\begin{aligned} C^\dagger u &= \nu P^\dagger^{-1} \lambda, \\ \Delta^\dagger \Delta &= \lambda \lambda^\dagger + P^\dagger \nu^{-1} P = 1_2 f^{-1}. \end{aligned} \quad (2.15)$$

The results (2.13) and (2.15) achieve a fairly explicit form for the gauge field $\tilde{A}_\mu$. To determine the corresponding form of the current J_μ we note that following form

(2.9):

$$\delta A_\mu = U^{-1} \delta \tilde{A}_\mu U + \mathcal{D}_\mu(A) U^{-1} \delta U ,$$

and

$$\int d^4x \operatorname{tr} \{ \delta A_\mu \tilde{J}_\mu \} = \int \operatorname{tr} \{ \delta \tilde{A}_\mu \tilde{J}_\mu \} ,$$

if $\mathcal{D}_\mu(A) J_\mu = 0$ and $\tilde{J}_\mu = U J_\mu U^{-1} (\mathcal{D}_\mu X = \partial_\mu X + [A_\mu, X])$. Since the current J_μ defined in (2.4) is covariantly conserved then we may use (2.2) with $A_\mu \rightarrow \tilde{A}_\mu$, $J_\mu \rightarrow \tilde{J}_\mu$ given by

$$\tilde{J}_\mu = \lambda^\dagger f P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f C^\dagger u . \quad (2.16)$$

In order for the variation of the determinant to be expressed entirely in terms of square matrices acting on the space $\mathcal{W}$ a further step is necessary. For this it is convenient to transform from the initial gauge field A_μ to a gauge field $\hat{A}_\mu$ and current $\hat{J}_\mu$, for the space $\mathcal{W}$ by defining

$$\hat{A}_\mu = a_\mu \lambda \lambda^\dagger , \quad \hat{J}_\mu = j_\mu \lambda \lambda^\dagger , \quad (2.17)$$

where

$$\tilde{A}_\mu = \lambda^\dagger a_\mu \lambda , \quad \tilde{J}_\mu = \lambda^\dagger j_\mu \lambda . \quad (2.18)$$

Using standard manipulations in the context of the general ADHM solution

$$\partial_\mu \{ f (e_\mu P^\dagger - P \bar{e}_\mu) f \} = 0 , \quad (2.19)$$

and hence

$$\partial_\mu j_\mu = -a_\mu \lambda \lambda^\dagger j_\mu + j_\mu \lambda \lambda^\dagger a_\mu , \quad (2.20)$$

which is sufficient to entail not only $\mathcal{D}_\mu(\tilde{A}) \tilde{J}_\mu = 0$ but also $\mathcal{D}_\mu(\hat{A}) \hat{J}_\mu = 0$. Furthermore, then

$$\begin{aligned} & \int d^4x \operatorname{tr} \{ \delta \tilde{A}_\mu \tilde{J}_\mu \} - \int d^4x \operatorname{tr} \{ \delta \hat{A}_\mu \hat{J}_\mu \} \\ &= \int d^4x \operatorname{tr} \{ \lambda \delta \lambda^\dagger (a_\mu \lambda \lambda^\dagger j_\mu - j_\mu \lambda \lambda^\dagger a_\mu) \} \\ &= 0 , \end{aligned} \quad (2.21)$$

by (2.20). There is no danger of surface terms contributing in (2.21) since asymptotically for $|x| \rightarrow \infty$ $a_\mu = O(x^{-3})$, $j_\mu = O(x^{-5})$. Thus the expression for the variation (2.2) becomes

$$\delta \zeta'(0) = \frac{1}{12\pi^2} \int d^4x \operatorname{tr}_{\mathcal{W}} \{ \delta \hat{A}_\mu \hat{J}_\mu \} , \quad (2.22)$$

with

$$\begin{aligned}\hat{A}_\mu &= -f\bar{e}_\mu\nu, \\ \hat{J}_\mu &= fP^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f\nu, \\ \nu &= C^\dagger u \lambda^\dagger = \nu P^{\dagger-1} \lambda \lambda^\dagger.\end{aligned}\quad (2.23)$$

The form for $\hat{A}_\mu$ in (2.23) is basically that given by Drinfeld and Manin [5] for the gauge group $SU(2k)$, or $Sp(k)$, and this expression has the virtue that the only rank conditions are on the matrix $\lambda\lambda^\dagger$ but they can be disregarded henceforth since λ and $\lambda^\dagger$ do not now appear separately.

In order to discuss the variation of $\hat{A}_\mu$ it is convenient to note from (2.23) and (2.15) that

$$\nu = \nu P^{\dagger-1} f^{-1} - P, \quad (2.24)$$

and in $\delta\hat{A}_\mu$ there are thus separate sets of terms arising from δf , δP or $\delta P^\dagger$ and $\delta\nu$ variations. Considering first the terms involving δf , their contributions to $\text{tr}\{\delta\hat{A}_\mu\hat{J}_\mu\}$ are

$$\begin{aligned}& -6 \text{tr}[\delta f \nu f P^\dagger f \nu P^{\dagger-1} f^{-1}] + 6 \text{tr}[\delta f \nu f P^\dagger f P] \\ & + \text{tr}[\bar{e}_\mu \nu P^{\dagger-1} f^{-1} \delta f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f \nu P^{\dagger-1}] \\ & - \text{tr}[f \bar{e}_\mu \nu P^{\dagger-1} f^{-1} \delta f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f P] \\ & + \text{tr}[\delta f \bar{e}_\mu P f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f \nu P^{\dagger-1} f^{-1}] \\ & - \text{tr}[\delta f \bar{e}_\mu P f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f P]\end{aligned}\quad (2.25)$$

(we use throughout the result that $e_\mu, \bar{e}_\mu$ can be commuted through f or ν and $\bar{e}_\mu P \bar{e}_\mu = -2P^\dagger$ etc.). Since

$$\partial_\mu P^{\dagger-1} = -P^{\dagger-1} \bar{e}_\mu \nu P^{\dagger-1},$$

and with

$$\begin{aligned}\partial_\mu P^\dagger &= \nu \bar{e}_\mu, \\ \partial_\mu (f^{-1} \delta f) 1_2 &= -f^{-1} \delta f (P^\dagger e_\mu + \bar{e}_\mu P) f \\ &\quad - \delta(P^\dagger e_\mu + \bar{e}_\mu P) f,\end{aligned}$$

the third term in (2.25) becomes

$$\begin{aligned}& -\partial_\mu \text{tr}[P^{\dagger-1} f^{-1} \delta f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f \nu] \\ & + 6 \text{tr}[P^{\dagger-1} f^{-1} \delta f \nu f P^\dagger f \nu] \\ & - \text{tr}[P^{\dagger-1} f^{-1} \delta f (P^\dagger e_\mu + \bar{e}_\mu P) f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f \nu] \\ & - \text{tr}[P^{\dagger-1} \delta(P^\dagger e_\mu + \bar{e}_\mu P) f P^\dagger f(e_\mu P^\dagger - P\bar{e}_\mu) f \nu].\end{aligned}\quad (2.26)$$

Thus (2.25) becomes

$$\begin{aligned}
& -\partial_\mu \operatorname{tr} [P^{\dagger-1} f^{-1} \delta f P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] \\
& + 6 \operatorname{tr} [\delta f \nu f P^\dagger f P] \\
& - \operatorname{tr} [\delta f P f P^\dagger (e_\mu P^\dagger f P \bar{e}_\mu + 2 P f P^\dagger)] \\
& - \operatorname{tr} [P^{\dagger-1} (\delta P^\dagger e_\mu + \bar{e}_\mu \delta P) f P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] .
\end{aligned} \tag{2.27}$$

Now considering the δP or $\delta P^\dagger$ contributions we have

$$\begin{aligned}
& \operatorname{tr} [f \bar{e}_\mu \nu P^{\dagger-1} \delta P^\dagger \delta P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f (\nu P^{\dagger-1} f^{-1} - P)] \\
& + \operatorname{tr} [f \bar{e}_\mu \delta P f P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f (\nu P^{\dagger-1} f^{-1} - P)] .
\end{aligned} \tag{2.28}$$

In a similar fashion as above but using (2.19),

$$\begin{aligned}
& \operatorname{tr} [P^{\dagger-1} \bar{e}_\mu \nu P^{\dagger-1} \delta P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] \\
& = -\partial_\mu \operatorname{tr} [P^{\dagger-1} \delta P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] \\
& + 6 \operatorname{tr} [P^{\dagger-1} \delta \nu f P^\dagger f \nu] ,
\end{aligned} \tag{2.29}$$

so combining (2.28) with the last term of (2.27) gives

$$\begin{aligned}
& -\partial_\mu \operatorname{tr} [P^{\dagger-1} \delta P^\dagger f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] \\
& + 6 \operatorname{tr} [P^{\dagger-1} \delta \nu f P^\dagger f \nu] \\
& - \operatorname{tr} [f \delta P f P^\dagger (e_\mu P^\dagger f P \bar{e}_\mu + 2 P f P^\dagger)] .
\end{aligned} \tag{2.30}$$

Finally the $\delta \nu$ contributions are

$$\begin{aligned}
& -6 \operatorname{tr} [\delta \nu f P^\dagger f \nu P^{\dagger-1}] \\
& + 6 \operatorname{tr} [f \delta \nu f P^\dagger f P] ,
\end{aligned} \tag{2.31}$$

and the result of the analysis is

$$\begin{aligned}
& \operatorname{tr} [\delta \hat{A}_\mu \hat{J}_\mu] \\
& = -\partial_\mu \operatorname{tr} [P^{\dagger-1} f^{-1} \delta (f P^\dagger) f (e_\mu P^\dagger - P \bar{e}_\mu) f \nu] \\
& + 6 \operatorname{tr} [\delta (f \nu) f P^\dagger f P] \\
& - \operatorname{tr} [\delta (f P) f P^\dagger f (e_\mu P^\dagger f P \bar{e}_\mu + 2 P f P^\dagger)] .
\end{aligned} \tag{2.32}$$

The derivative terms in (2.32) gives a vanishing contribution to (2.22) since although $f P^\dagger = O(x^{-1})$, $f \nu = O(x^{-2})$ asymptotically $\delta (f P^\dagger) = O(x^{-2})$. It is now convenient to introduce some further notation so defining

$$X = f P, \quad \bar{X} = f P^\dagger, \quad Y = f \nu 1_2, \tag{2.33}$$

we have the result

$$\begin{aligned} & \delta \zeta'(0) \\ &= \frac{1}{12\pi^2} \int d^4x \{6 \operatorname{tr}[\delta Y X \bar{X}] - \operatorname{tr}[\delta X \bar{X}(e_\mu \bar{X} X \bar{e}_\mu + 2X \bar{X})]\} . \end{aligned} \quad (2.34)$$

To proceed further it is necessary to find a form of the integrand in (2.34), by integrating by parts, so that variation can be taken outside the trace. The following derivative relations are of use to achieve this aim:

$$\begin{aligned} \partial_\mu \bar{X} e_\mu &= -4Y + \bar{e}_\mu X \bar{X} e_\mu - 2\bar{X} X , \\ \partial_\mu \bar{e}_\mu X &= -4Y + 2\bar{X} X , \\ \partial_\mu e_\mu \bar{X} X &= -4Y X + 2X Y + e_\mu \bar{X} X \bar{e}_\mu X , \\ \partial_\mu \bar{X} X \bar{e}_\mu &= -4\bar{X} Y + 2Y X + e_\mu X \bar{X} X e_\mu + 2\bar{X}(e_\mu \bar{X} X \bar{e}_\mu - X \bar{X}) . \end{aligned}$$

With these results we may show that

$$\begin{aligned} & 8 \operatorname{tr}[\delta Y \bar{X} X] \\ &= \partial_\mu \operatorname{tr}[(\delta \bar{X} e_\mu + \bar{e}_\mu \delta X) \bar{X} X] \\ &\quad - 4 \operatorname{tr}[\delta(X \bar{X}) Y] + 2 \operatorname{tr}[\delta(\bar{X} X) Y] \\ &\quad + 2 \operatorname{tr}[\delta(\bar{e}_\mu X \bar{X} e_\mu - \bar{X} X) X X] \\ &\quad + \operatorname{tr}[\delta X \bar{X} e_\mu \bar{X} X \bar{e}_\mu] , \end{aligned} \quad (2.35)$$

and as

$$\begin{aligned} \operatorname{tr}[\delta(X \bar{X}) Y] &= \operatorname{tr}[\delta(\bar{X} X) Y] , \\ \operatorname{tr}[\bar{e}_\mu \delta(X \bar{X}) e_\mu \bar{X} X] &= \operatorname{tr}[\bar{e}_\mu X \bar{X} e_\mu \delta(\bar{X} X)] , \end{aligned}$$

then

$$\begin{aligned} & 6 \operatorname{tr}[\delta Y X \bar{X}] - \operatorname{tr}[\delta X \bar{X} e_\mu \bar{X} X \bar{e}_\mu] \\ &= \partial_\mu \operatorname{tr}[(\delta \bar{X} e_\mu + \bar{e}_\mu \delta X) \bar{X} X] \\ &\quad - 2\delta \operatorname{tr}[Y X \bar{X}] \\ &\quad + \delta \operatorname{tr}[(\bar{e}_\mu X \bar{X} e_\mu - \bar{X} X) \bar{X} X] . \end{aligned} \quad (2.36)$$

The remaining term in (2.34) can be written as

$$\begin{aligned} \operatorname{tr}[\delta X \bar{X} X \bar{X}] &= \frac{1}{4} \delta \operatorname{tr}[X \bar{X} X \bar{X}] \\ &\quad + \frac{1}{2} \operatorname{tr}[(\delta X \bar{X} - X \delta \bar{X}) X \bar{X}] , \end{aligned} \quad (2.37)$$

so combining (2.36) and (2.37) in (2.34) gives

$$\begin{aligned} \delta \zeta'(0) = & \frac{1}{12\pi^2} \int d^4x \delta \{ -2 \operatorname{tr}[YX\bar{X}] + \operatorname{tr}[\bar{e}_\mu X \bar{X} e_\mu \bar{X} X] - \frac{3}{2} \operatorname{tr}[\bar{X} X \bar{X} X] \} \\ & + \delta\theta, \end{aligned} \quad (2.38)$$

where

$$\delta\theta = -\frac{1}{12\pi^2} \int d^4x \operatorname{tr}[(\delta X \bar{X} - X \delta \bar{X}) X \bar{X}], \quad (2.39)$$

since the derivative term in (2.36) gives a vanishing surface contribution. For the present we do not discuss the $\delta\theta$ term, analysing it further in the conclusion. The remainder of (2.38) may be simplified with the aid of

$$\begin{aligned} \partial^2 \operatorname{tr}[Y] &= -8 \operatorname{tr}[YY] + 8 \operatorname{tr}[YX\bar{X}], \\ \partial^2 \operatorname{tr}[\bar{X}X] &= 8 \operatorname{tr}[YY] - 32 \operatorname{tr}[YX\bar{X}] + 8 \operatorname{tr}[(\bar{e}_\mu X \bar{X} e_\mu - \bar{X}X) \bar{X}X], \end{aligned} \quad (2.40)$$

which subsumes the result of ref. [1]

$$\partial^2 T = \frac{1}{2} \operatorname{tr}(F_{\mu\nu} F_{\mu\nu}), \quad T = \operatorname{tr}[2Y - \bar{X}X]. \quad (2.41)$$

While the integral (2.38) is finite, the variation cannot be taken outside the integration without a potential divergence as

$$\operatorname{tr}[YY], \quad \operatorname{tr}[YX\bar{X}], \quad \operatorname{tr}[\bar{X}X\bar{X}X], \quad \frac{1}{4} \operatorname{tr}[\bar{e}_\mu X \bar{X} e_\mu \bar{X}X],$$

each behave asymptotically as $2k(x^2)^{-2}$. However this divergence can be regularised by hand to write

$$\begin{aligned} \zeta'(0) = & \lim_{R^2 \rightarrow \infty} \left\{ \frac{1}{12\pi^2} \int_{x^2 < R^2} d^4x (\operatorname{tr}[YY] - \frac{1}{2} \operatorname{tr}[\bar{X}X\bar{X}X]) - \frac{1}{12} k \ln R^2 \right\} \\ & + 0 + \alpha(k), \end{aligned} \quad (2.42)$$

where $\alpha(k)$ is independent of the parameters of the general instanton solution. It is reasonable to conjecture, for flat space, that it is linear in k (any change in the form of the integrand using (2.40) just introduces a constant term proportional to k , the term independent of k , as shown in ref. [1], contains the infrared divergence which arises when $\zeta'(0)$ is considered on flat space). Since $f(x)$, defined in (2.11), is regular everywhere there are no other singularities in the integral expression (2.42) which thus gives $-\ln \det(-D^2)$ up to any additional contribution coming from the unanalysed $\delta\theta$ term.

3. 't Hooft solution

In this section we check the result (2.42) by calculating it for the 't Hooft solution which provides an explicit parametrisation of the ADHM construction for the gauge group $SU(2)$. It is convenient to re-express (2.42) using (2.40), and disregard the divergence difficulties, in the form

$$\delta \zeta'(0) = \frac{1}{12\pi^2} \delta \int d^4x \left\{ \frac{5}{2} \text{tr}[YY] - \frac{1}{2} \text{tr}[\bar{e}_\mu X \bar{X} \bar{e}_\mu \bar{X} X] \right\} + \delta \theta. \quad (3.1)$$

The quaternionic trace may be evaluated using

$$\begin{aligned} Y &= f\nu 1_2, & X &= \frac{1}{2}e \cdot k, & \bar{X} &= \frac{1}{2}\bar{e} \cdot k, \\ k_\alpha &= f\partial_\alpha f^{-1}, \end{aligned} \quad (3.2)$$

to give

$$\delta \zeta'(0) = \frac{1}{12\pi^2} \delta \int d^4x \left\{ 5 \text{tr}[f\nu f\nu] - \frac{1}{4} \text{tr}[k^2 k^2] \right\} + \delta \theta, \quad (3.3)$$

where now the trace is over the k -dimensional space W .

The basic field in the 't Hooft solution is

$$\varphi = \sum_0^k \frac{\lambda_i^2}{x_i^2}, \quad x_i = x - y_i \quad (3.4)$$

and following ref. (1), in terms of the matrices

$$X = \begin{bmatrix} x_1 & & 0 \\ & \ddots & \\ 0 & & x_k \end{bmatrix}, \quad \lambda = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{bmatrix}, \quad (3.5)$$

we find

$$\begin{aligned} f^{-1} &= X^2 \lambda_0^2 + x_0^2 \lambda \lambda^T, \\ \nu &= \lambda_0^2 1_k + \lambda \lambda^T. \end{aligned} \quad (3.6)$$

The inversion of f^{-1} is easily found

$$f = (X^2)^{-1} \lambda_0^{-2} - \frac{1}{\varphi} (X^2)^{-1} \frac{\lambda \lambda^T}{\lambda_0^2} (X^2)^{-1}, \quad (3.7)$$

so $f\nu$ and $k_\alpha = f\partial_\alpha f^{-1}$ can be calculated as

$$f\nu = (X^2)^{-1} - \frac{1}{\varphi} (X^2)^{-1} \lambda \lambda^T \left((X^2)^{-1} - \frac{1}{x_0^2} 1_k \right), \quad (3.8)$$

$$\frac{1}{2}k_\alpha = (X^2)^{-1}X_\alpha - \frac{1}{\varphi}(X^2)^{-1}\lambda\lambda^\top\left((X^2)^{-1}X_\alpha - \frac{x_{0\alpha}}{x_0^2}1_k\right).$$

It is then straightforward to evaluate

$$\begin{aligned}\mathrm{tr}[f\nu f\nu] &= \sum_0^k \frac{1}{x_i^4} - 2\frac{\varphi^{(3)}}{\varphi} + \frac{\varphi^{(2)2}}{\varphi^2}, \\ \frac{1}{16}\mathrm{tr}[k^2k^2] &= \sum_0^k \frac{1}{x_i^4} - 4\frac{\varphi^{(3)}}{\varphi} + \frac{1}{2}\frac{\partial\varphi \cdot \partial\varphi^{(2)}}{\varphi^2} + \frac{\varphi^{(2)2}}{\varphi^2} \\ &\quad + \frac{1}{16}\frac{(\partial\varphi \cdot \partial\varphi)^2}{\varphi^4} - \frac{1}{2}\frac{\partial\varphi \cdot \partial\varphi}{\varphi^3}\varphi^{(2)} \\ &\quad + \frac{1}{\varphi^2}\pi_{\alpha\beta}\pi_{\alpha\beta} - \frac{1}{2}\frac{\partial_\alpha\varphi\partial_\beta\varphi\pi_{\alpha\beta}}{\varphi^3},\end{aligned}\tag{3.9}$$

where

$$\varphi^{(n)} = \sum_0^k \frac{\lambda_i^2}{(x_i^2)^n}, \quad \pi_{\alpha\beta} = \sum_0^k \lambda_i^2 \frac{x_{i\alpha}x_{i\beta}}{(x_i^2)^3}.$$

Using (3.9) and

$$\pi_{\alpha\beta} = \frac{1}{8}\partial_\alpha\partial_\beta\varphi + \frac{1}{4}\delta_{\alpha\beta}\varphi^{(2)}, \quad \varphi^{(3)} = \frac{1}{8}\partial^2\varphi^{(2)},$$

we obtain

$$\begin{aligned}5\mathrm{tr}[f\nu f\nu] - \frac{1}{4}\mathrm{tr}[k^2k^2] &= \sum_0^k \frac{1}{x_i^4} - \frac{1}{16}\frac{(\partial\varphi \cdot \partial\varphi)^2}{\varphi^4} + \pi^2 \sum_0^k \delta^4(x_i) \\ &\quad + \partial_\alpha \left\{ \frac{3}{4}\frac{1}{\varphi}\partial_\alpha\varphi^{(2)} - \frac{5}{4}\frac{\partial_\alpha\varphi}{\varphi^2}\varphi^{(2)} - \frac{1}{32}\frac{1}{\varphi^2}\partial_\alpha(\partial\varphi \cdot \partial\varphi) + \frac{1}{16}\partial_\alpha\varphi\frac{\partial\varphi \cdot \partial\varphi}{\varphi^3} \right\}.\end{aligned}\tag{3.10}$$

In (3.10) the $\delta^4(x_i)$ terms cancel similar terms arising from $-\frac{1}{4}\partial_\alpha(x_{i\alpha}/x_i^4)$. Inserting (3.10) in (3.3) we may discard the derivative term to find

$$\delta\zeta'(0) = \frac{1}{12\pi^2}\delta\int d^4x \left\{ \sum_0^k \frac{1}{x_i^4} - \frac{1}{16}\frac{(\partial\varphi \cdot \partial\varphi)^2}{\varphi^4} \right\},\tag{3.11}$$

since the $\delta\theta$ term vanishes in this case. Eq. (3.11) essentially agrees with the result of Brown and Creamer [2]. The first term in the integrand serves to remove all the singularities at $x_i \rightarrow 0$ (since our starting point (3.1) is regular for all finite x) but at

the expense of ensuring that the integrand behaves as k/x^4 for $x \rightarrow \infty$. By itself it may be discarded since any sensible regularisation ensures that $\int d^4x x_i^{-4}$ is defined to be zero.

4. Conformal properties

It is of some interest to check that the result (2.42) has the right properties under conformal transformation to agree with the general analysis of ref. [1]. A conformal transformation expressed in quaternionic form is

$$x \rightarrow x' = (\alpha x + \beta)(\gamma x + \chi)^{-1}, \quad (4.1)$$

which implies

$$dx \rightarrow dx' = \Omega(x) \overline{R'(x)} dx \overline{R(x)}, \quad (4.2)$$

where

$$\Omega(x) = \frac{|\Lambda||\gamma|}{|\gamma x + \chi|^2}, \quad R(x) = \frac{\gamma x + \chi}{|\gamma x + \chi|}, \quad R'(x) = \frac{\bar{\gamma}(\gamma x + \chi) \bar{\Lambda}}{|\gamma||\Lambda||\gamma x + \chi|},$$

$$\Lambda = \alpha \gamma^{-1} \chi - \beta.$$

In (4.2) $|R'| = |R| = 1$ so R, R' represent local rotations on the quaternionic space and Ω is the local scale change $|dx'|^2 = \Omega(x)^2 |dx|^2$. Under the action of the transformation (4.1) there is induced a change in the parameters of the gauge field A_μ as represented in the matrix

$$\Delta(x') = \Delta'(x)(\gamma x + \chi)^{-1}, \quad (a' \ b') = (a \ b) \begin{pmatrix} \chi & \gamma \\ \beta & \alpha \end{pmatrix}. \quad (4.3)$$

As a consequence of (4.3) we find

$$f(x') = f'(x)|\gamma x + \chi|^2, \\ b = b' \gamma^{-1}(\gamma x + \chi) \Lambda^{-1} - \Delta'(x) \Lambda^{-1}. \quad (4.4)$$

Using (4.3) and (4.4) we find under $x \rightarrow x'(k_\mu = f \partial_\mu f^{-1})$

$$f\nu \rightarrow \Omega^{-2} f' \nu' - \frac{1}{2} \Omega^{-1} k' \cdot \partial \Omega^{-1} + \frac{1}{8} \Omega^{-1} \partial^2 \Omega^{-1}, \\ \bar{X} \rightarrow \Omega^{-1} R \bar{X}' R' - R \frac{1}{2} \bar{e} \cdot \partial \Omega^{-1} R', \\ \bar{X} X \rightarrow \Omega^{-2} R \bar{X}' X' \bar{R} - \frac{1}{2} \Omega^{-1} k' \cdot \partial \Omega^{-1} 1_2 + \frac{1}{4} \partial \Omega^{-1} \cdot \partial \Omega^{-1} 1_2. \quad (4.5)$$

Since from the explicit form of Ω , $\partial^2 \Omega^{-1} = 2\Omega \partial \Omega^{-1} \cdot \partial \Omega^{-1}$, the inhomogeneous terms in the conformal transformation of $\bar{X} X$, or $X \bar{X}$, and $Y = f \nu 1_2$ are identical.

The results (4.4) allow the conformal properties of the integrals that appear in (2.42) to be determined. The transformation is induced by performing the trans-

formation $x \rightarrow x'$ with $d^4x' = \Omega^4 d^4x$. However, it is necessary to be careful with the consequences of regularisation where

$$\int_{\text{reg}} d^4x \operatorname{tr}[YY] = \lim_{R^2 \rightarrow \infty} \left\{ \int_{x^2 < R^2} d^4x \operatorname{tr}[YY] - 2\pi^2 k \ln R^2 \right\}. \quad (4.6)$$

For an infinitesimal conformal transformation $\Omega = 1 - \omega$, with $\omega(x) = \sigma + \tau_\mu x_\mu$, as in ref. [1], and asymptotically for large x from (4.1)

$$\begin{aligned} |x'|^2 &\sim (1 - 2\sigma)|x|^2 - |x|^2 \tau_\mu x_\mu, \\ \theta(R^2 - |x'|^2) &\sim \theta(R^2 - |x|^2) + \delta(R^2 - |x|^2)(2\sigma R^2 + R^3 \tau_\mu x_\mu), \end{aligned} \quad (4.7)$$

if $\hat{x} = x/|x|$. Hence if δ_c denotes an infinitesimal change in the parameters as given in (4.3)

$$\begin{aligned} \delta_c \int_{\text{reg}} d^4x \operatorname{tr}[YY] &= \int d^4x \partial_\mu \omega \operatorname{tr}[k_\mu Y] \\ &- \lim_{R^2 \rightarrow \infty} \int d^4x \delta(R^2 - |x|^2)(2\sigma R^2 + R^3 \tau_\mu x_\mu) \operatorname{tr}[YY]. \end{aligned} \quad (4.8)$$

This can be evaluated using $\operatorname{tr}[k_\mu Y] = -\partial_\mu \operatorname{tr}[Y]$ and the asymptotic form

$$Y \sim 1_2 \left\{ \frac{1}{|x|^2} - \nu^{-1} \frac{2\rho_\mu x_\mu}{|x|^4} \right\}, \quad b^\dagger a = \rho \cdot e, \quad (4.9)$$

Hence

$$\delta_c \int d^4x \operatorname{tr}[YY] = -4\pi^2(\sigma k - \operatorname{tr}_W [\nu^{-1} \rho \cdot \tau]). \quad (4.10)$$

Since $X\bar{X}$ has an identical asymptotic form as Y in (4.9) and

$$\operatorname{tr}[k_\mu X\bar{X}] = -\frac{1}{2} \partial_\mu \operatorname{tr}[Y] - \frac{1}{2} \partial_\mu \operatorname{tr}[X\bar{X}],$$

then it follows that the corresponding integrals over $\operatorname{tr}[YX\bar{X}]$, $\operatorname{tr}[X\bar{X}X\bar{X}]$, $\frac{1}{4} \operatorname{tr}[X\bar{X}e_\mu X\bar{X}e_\mu]$ have identical conformal transformation properties as in (4.10). Furthermore, from (2.42), assuming $\delta_c \theta = 0$, as we shall show later,

$$\delta_c \zeta'(0) = -\frac{1}{6} \sigma k + \frac{1}{6} \operatorname{tr}_W [\nu^{-1} \rho \cdot \tau], \quad (4.11)$$

in accord with ref. [1].

Obviously by taking linear combinations of the four integrals discussed above it is possible to form conformal invariants, but the combinations given in (2.40) obviously result in trivial constants. The one non-trivial expression can also be seen by noting that under $x \rightarrow x'$

$$V^\dagger b f b^\dagger V \rightarrow \Omega^{-2} V'^\dagger b' f' b'^\dagger V',$$

so

$$\begin{aligned} & \int d^4x \operatorname{tr} [V^\dagger b f b^\dagger V V^\dagger b f b^\dagger V] \\ &= \int d^4x \{ \operatorname{tr} [Y Y] - 2 \operatorname{tr} [Y X \bar{X}] + \operatorname{tr} [X \bar{X} X X] \} \end{aligned} \quad (4.12)$$

is a conformal invariant.

5. Conclusion

The results presented here may perhaps be a small step towards the complete evaluation of multi-instanton determinants and a realisation of the ideas discussed in refs. [1,4]. However, there remains the term $\delta\theta$ which was found in (2.38) and (2.39). We have been unable either to show that it is zero or to take the variation outside the trace. If the quaternionic trace is evaluated, then

$$\begin{aligned} & \operatorname{tr}_{\mathcal{W}} [(\delta X \bar{X} - X \delta \bar{X}) X \bar{X}] \\ &= -\epsilon_{\alpha\beta\gamma\delta} \operatorname{tr}_{\mathcal{W}} [\delta k_\alpha k_\beta k_\gamma k_\delta], \end{aligned} \quad (5.1)$$

so this contribution uniquely involves the 4-dimensional symbol. By cyclicity of the trace

$$\epsilon_{\alpha\beta\gamma\delta} \operatorname{tr} [k_\alpha k_\beta k_\gamma k_\delta] = 0.$$

From the definition $k_\alpha = f \partial_\alpha f^{-1}$ then

$$\begin{aligned} \partial_\alpha k_\beta &= -k_\alpha k_\beta + 2f\nu\delta_{\alpha\beta}, \\ \delta k_\alpha &= -[\delta g, k_\alpha] + \partial_\alpha \delta g, \\ \delta g &= f\delta f^{-1}. \end{aligned} \quad (5.2)$$

Hence (5.2) allows us, by discarding a derivative, to obtain the manifestly real form

$$\delta\theta = \frac{1}{12\pi^2} \int d^4x \epsilon_{\alpha\beta\gamma\delta} \operatorname{tr} [\delta g k_\alpha k_\beta k_\gamma k_\delta] \quad (5.3)$$

but clearly further insight is needed.

For a conformal change in the parameters (4.3) then from (4.4) we may easily see that

$$f\delta_c f^{-1} = r 1_k + s_\mu k_\mu$$

and as $\epsilon_{\alpha\beta\gamma\delta} \operatorname{tr} [k_\mu k_\alpha k_\beta k_\gamma k_\delta]$ is totally antisymmetric in $k_\mu, k_\alpha, k_\beta, k_\gamma, k_\delta$ and thus vanishes $\delta_c \theta = 0$ so this term is a conformal invariant.

Apart from this it would be nice to relate the present results to the conjectures in ref. [4] about the form of the final result. For the extended 't Hooft solution

Belavin et al., obtain the finite expression (with some corrections)

$$\begin{aligned} \zeta'(0) = & -\frac{1}{12 \times 16\pi^2} \int d^4x \ln \det f \nu \partial^2 \partial^2 \ln \det f \nu \\ & - \frac{1}{12} \sum_{i \neq j} \ln(\nu_i - \nu_j)^2 - \frac{1}{6} \sum_0^k \ln \lambda_i^2 \\ & + \frac{k+1}{6} \ln \sum_0^k \lambda_i^2 + \text{constant}. \end{aligned} \quad (5.4)$$

One result which is perhaps of interest in relation to this is that

$$\begin{aligned} T = & -\frac{1}{2} \partial^2 \ln \det f, \\ & - \partial^2 \partial^2 \ln \det f = \text{tr}[F_{\mu\nu} F_{\mu\nu}], \end{aligned} \quad (5.5)$$

which can easily be shown using the result for T in (2.41).

The formula (5.5) allows one to see an intriguing parallel between (5.4) and the corresponding results in 2 dimensional models. Perhaps to derive a result akin to (5.4) more generally requires an understanding of the relationship of determinants for different representations of the gauge group.

Note added

Recently Berg and Luscher (DESY preprint) have completed a calculation similar to that undertaken here but for the Dirac operator. Furthermore they have shown how to handle the θ term whose variation is here given in (5.3). If we introduce a variable $x_5 = s$, in addition to the usual x , and define

$$\begin{aligned} S = & \epsilon_{\alpha\beta\gamma\delta\rho} \text{tr}[m_\alpha m_\beta m_\gamma m_\delta m_\rho], \\ m_\alpha = & l^{-1} \partial_\alpha l, \quad \epsilon_{\alpha\beta\gamma\delta 5} = \epsilon_{\alpha\beta\gamma\delta}, \end{aligned}$$

then using results analogous to (5.2)

$$\delta S = 5 \epsilon_{\alpha\beta\gamma\delta\rho} \partial_\rho \text{tr}[m_\alpha m_\beta m_\gamma m_\delta l^{-1} \delta l].$$

By integrating this we may reproduce (5.3) if we define

$$\theta = \frac{1}{12\pi^2} \frac{1}{s} \int d^4x \int_0^1 ds S,$$

$$l(x, s) = f(x)^{-1} s + f_0(x)^{-1} (1 - s)$$

where in $\delta\theta$ the only contributions from the integration of δS are surface terms arising from $s = 1$, which reproduces (5.3), and $s = 0$, where for some trivial choice of $f_0(x)$ (e.g. $1_k(1 + x^2)^{-1}$) the result is zero.

By combining our results with those of ref. [1] and the above we have for the

determinant on S^4 for radius a

$$\zeta'_{D2}(0) = \frac{1}{48\pi^2} \int_{\text{reg}} d^4x \{20 \operatorname{tr}[f\nu f\nu] - \operatorname{tr}[k^2 k^2]\} + \theta$$

$$- \frac{1}{96\pi^2} \int d^4x \ln\left(1 + \frac{x^2}{a^2}\right) \operatorname{tr}[F_{\mu\nu} F_{\mu\nu}] - \frac{N}{90} \ln a^2 + \alpha(k) ,$$

where $N = 2r$, n for gauge group $\operatorname{Sp}(r)$, $\operatorname{SU}(n)$.

Berg and Luscher also show that $\alpha(k)$ is linear in k as we conjectured. For $N > 2k$ the general instanton solution can be achieved by embedding the solution for $\operatorname{Sp}(k)$ or $\operatorname{SU}(2k)$ while for $N < 2k$ the general instanton solution is a limiting boundary surface in the connected parameter space of the solution for $\operatorname{Sp}(k)$ or $\operatorname{SU}(2k)$. Thus it suffices to consider just the determinant for $\operatorname{Sp}(k)$ or $\operatorname{SU}(2k)$. It may then be evaluated exactly by considering the special configuration formed by a superposition of k single $\operatorname{Sp}(1) = \operatorname{SU}(2)$ instantons belonging to k commuting $\operatorname{Sp}(1)$ or $\operatorname{SU}(2)$ subgroups, each centered at $x = 0$ with a scale a so that f can be taken to be of the form $f(x) = 1_k(x^2 + a^2)^{-1}$. The eigenvalues of the conformal scalar operator $-D^2 + \frac{1}{6}R$ may be found explicitly in this case, being $1/a^2 n(n+2)$ with degeneracy $k \frac{2}{3}n(n+1)(n+2)$, $n = 1, 2, \dots$. By comparing with the result when $k = 0$ and the eigenvalues are $1/a^2 n(n+1)$ with degeneracy $N \frac{1}{6}n(n+1)(2n+1)$ and using the zeta function method of regularisation we obtain in general

$$\alpha(k) = N\alpha_0 + \alpha_1 k ,$$

$$\alpha_0 = \frac{1}{3}\zeta'_R(-3) + \frac{1}{3}\zeta'_R(-1) + \frac{1}{144} ,$$

$$\alpha_1 = 2\zeta'_R(-1) - \frac{1}{8}$$

for $\zeta_R(s)$ the standard Riemann zeta function.

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